

BRAIN-COMPUTER INTERFACE BASED CONTROL FOR DISABLED

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Abstract— Brain-computer interface (BCI) has gained popularity in recent years as an alternate communication mechanism between the human brain and technology for output. WHO accounts for 15% of the world's population with disabilities, and there is a critical need for assistive systems to aid those persons in carrying out their everyday activities. This study can help disabled people to increase their independence and quality of life. Electrical impulses from the human scalp are detected and converted into control commands that may be utilized to control external equipment, computer screens, and virtual objects in real time. The present work uses the Deep Learning method to interpret motor imagery signals for Brain-Computer Interface applications. ERD patterns are also gathered that represent the activation and deactivation of the motor cortex. In addition, task-related band power changes are recorded in every trial, which can show stimulation of sensory-motor areas. Furthermore, the goal is to categorize EEG data derived from a Motor Imagery task into two classes, viz. left and right, using the convolutional neural networks. The time-frequency pictures of EEG data fed as input to the convolutional neural network were obtained using continuous wavelet transform. The three channels were separated using data segmentation to convert the raw data into images. To get over the issue of few trials in motor imagery tasks, transfer learning strategies are employed. Transfer learning is applied to three premade neural networks GoogLeNet, AlexNet, and ResNet50, to proceed with CNN classification. Upon analyzing results, AlexNet achieved the best accuracy among the others.

Keywords—Brain-Computer interface (BCI), Assistive Devices, Convolutional Neural Network (CNN), Electroencephalogram (EEG), Feature extraction, Classification.

I. INTRODUCTION

Brain-computer interface (BCI) has been investigated with the goal of developing assistive technology for persons with severe mobility impairments. Locked-in syndrome (LIS) is a neurological condition induced by neurodegenerative illnesses such as Amyotrophic Lateral Sclerosis (ALS) or stroke. As a result, it may be able to assist people in communicating with the outside world without the use of limbs or muscles. Wearable lower-limb (LL) exoskeletons, orthoses, prostheses, wheelchairs, and assistive-robot technologies might all be beneficial to patients. In BCI technology, collecting EEG data characteristics is crucial. By monitoring and analyzing human physiological signals, the

BCI can operate external devices, for example, control of cursors and wheelchairs. Thus, BCI may assist people in communicating with the outdoor world, lacking the use of limbs or muscles. The recording of electroencephalograph (EEG) during motor imagery tasks is a common BCI paradigm [1]. The neuronal underpinning of motor imagery (MI), i.e., picturing motion of specific regions of the body, is the phenomenon of Event-Related Synchronization (ERS) and Event-Related Desynchronization (ERD). When humans perform particular MI tasks, energy levels rise in the contralateral region while falling levels in the ipsilateral., allowing for the acquisition and recognition of various signals, which are then translated into directives [2]. Research have shown that deep learning approaches can be used to solve difficulties in specialized industries. In the fields of image identification and natural language processing, deep convolutional neural networks have made significant progress. When recognizing EEGs using the CNN technique, the network receives two different types of input. The first technique involves the utilization of raw EEG data. The CNN model is capable of learning characteristics from the hidden layers on its own and is used as a classifier in another pattern, extracting features in advance using methods such as Continuous wavelet transform (CWT) or Short-Time Fourier Transform (STFT), and the features which are used to train the network [3].

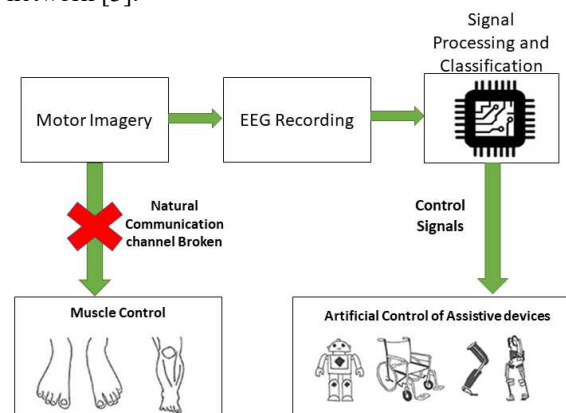


Figure 1 Motor Imagery framework

Deep learning models are made up of several inference layers that use data representations with different degrees of abstraction and are currently gaining popularity. As a result,

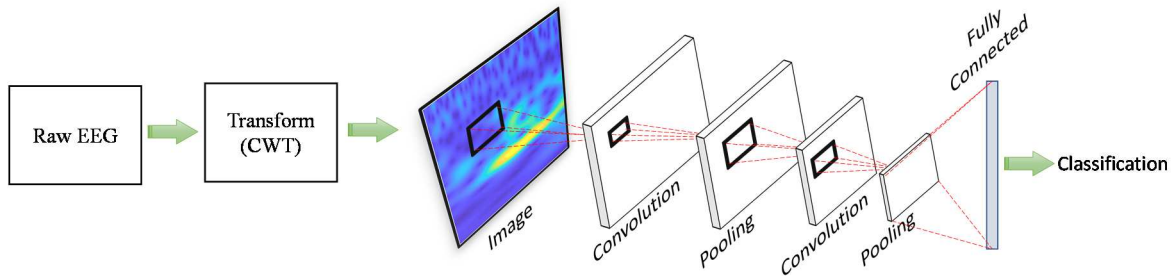


Figure 2 Deep learning method adopted in the study

CNN models can be used to learn aspects related to brain imaging and neuroscience research. The computational load is increased by the huge number of hyperparameters to be learned. Consequently, it is inappropriate for online processing and complex multilayer integration to encode important characteristics of the incoming EEG data at every abstraction level. Deep learning algorithms require a large quantity of knowledge in order to provide trustworthy results; As a result, huge data sets are used to feed information into the system. Artificial neural networks then are ready to classify information based on the answers to a series of binary yes/no questions. In deep CNN, data processing involves extremely complex mathematical computations. Face recognition software, for example, learns to detect and distinguish edges and features on faces from more significant sections of faces, and finally, general representations of faces. The algorithm improves its own training over time, increasing the likelihood of the right replies. In this situation, the facial recognition program will recognize faces properly over time. [3]. Because of its discriminative qualities and the low cost of signal gathering, MI is often used in BCI systems. Furthermore, after a stroke, MI data is commonly utilized to augment rehabilitative treatment. Despite this, it is challenging to process MI data. Moreover, the majority of processing and classification methods are complicated. Many systems are being hampered by poor classification accuracy due to the instabilities of EEG signals. Furthermore, many classifiers overlook time-series data, despite the fact that including it improves classification accuracy [4]. Figure 1 shows the framework adopted in this work.

II. DATA ACQUISITION

In this paper, BCI competition IV dataset 2b is used for the classification of MI signals. This data set contains EEG data from nine different people. During the recording, all of the participants were seated in an armchair, facing a computer display that was roughly 1 meter away and at eye level. Five sessions are offered for each subject, with the first two sessions being training data without feedback (screening) and the last three sessions containing feedback. The MI of the left hand (class 1) and the right hand (class 2) were the two classes in the cue-based screening paradigm. Within two weeks, each subject had two screening sessions with no feedback recorded on two distinct days. Each session had six runs with ten trials each and two imaging lessons. There were 20 trials per run and 120 trials per session as a result of this. For each person, a total of 120 repetitions of each MI class were provided.

Table 1 Details of the dataset events used in this study

Channels	Subjects	Trials	Sampling	MI Class
C3, Cz, C4	9	400	250	2 (left/Right)

III. METHODOLOGY

In most cases, there are three main processes for interpreting EEG signals. Ineffectual components are removed during the pre-processing step, followed by the feature extraction stage in which discriminating features from different classes are acquired. Finally, in the classification step, a model is created to map characteristics to their categories and then utilizes that model to predict future signals. Figure 2 depicts the EEG-based BCI for MI applications adopted in this work. EEG signals acquired from the three channels were digitized at a sampling rate of 250Hz [9,10]. EEG data was pre-processed before being fed to remove noise and artifacts. In this manner, the most important elements for categorization can then be selected from the retrieved features from the EEG data. Based on the collected characteristics, the classifier decides which motor action is envisioned. The data was divided into 5-second-long windows. Noise like power lines and baseline noise from EEG signals have been removed by applying Butterworth band-filter and Notch filter, respectively.

A. DATA SEGMENTATION

In our work presented, the first step is to extract the data of the particular event, i.e., left and right. Around 1250 samples are extracted for 5 sec long data to perform feature selection. Each session consisted of six runs of ten trials each, as well as two classes of imagery. As a result, each run had 20 trials, and each session had 120 trials. Each person got access to data from 120 repeats of each MI class.

B. FEATURE EXTRACTION

Feature extraction is critical in brain-computer interface applications because raw EEG data are nonstationary signals that are damaged by noise or artifacts present in the environment when they are being collected; nevertheless, they still contain useful information. [5]. The dimensionality of data is also reduced, which reduces computational complexity. This strategy is critical for improving the BCI system's classification accuracy. The EEG signal is a nonstationary time-domain signal, and the essential information, such as signal energy, is examined as a function of time or frequency afterward; suitable statistic measures are

created to explain the signal's features correctly. Fast Fourier Transform (FFT) Method, Wavelet Transform (WT) Method, and Discrete Wavelet Transform (DWT) are some of the most often utilized feature extraction techniques [6].

1. Motor Imagery EEG Image Formation Using Continuous Wavelet Transform

By collecting MI characteristics that appear in a certain waveband during the MI task, we created a substitute two-dimensional image. The final picture was created using the time-frequency representation of the MI EEG data. As a result of a resolution trade-off between time and frequency, STFT, extensively utilized for the depiction of the temporal frequency, MI EEG signal cannot be understood. When the window in STFT is too small, it causes distortion on low frequency. To overcome this issue, we developed a picture of the EEG signal using the continuous wavelet transform (CWT) [11]. When compared to Fourier transform (FT) based techniques, the CWT method creates a complete depiction of motor imagery-related EEG signals. Furthermore, using a one-dimensional convolutional neural network (CNN), it can distinguish between temporal changes in EEG data from motor imagery tasks depending on specific frequency and electrode. The use of wavelet transforms and neural networks in combination might lead to the creation of sophisticated EEG signal processing and machine learning techniques for employment in the development of assistive technologies for the disabled [19,21].

The procedures used by the CWT and the FT are quite similar. The initial signal and a sinusoidal signal are correlated using FT. In the same way, the correlation coefficients between the original signal and a mother wavelet are obtained using CWT. CWT, on the other hand, assigns a time-frequency domain to a signal by manipulating the form of the mother wavelet, unlike the FT, which decomposes the signal into a frequency domain. Scaling and shifting parameters influence the shape of the mother wavelet in this case.

$$CWT(\omega, s) = \frac{1}{\sqrt{|s|}} \int x(t) \varphi\left(\frac{t - \omega}{s}\right) dt \quad (1)$$

In this work, $x(t)$ indicates the MI EEG signal, φ is a wavelet mother indicating a time change/translation parameter, while s refers to a scaling parameter. The correlation coefficients of CWT are represented by equation (1). The images extracted using CWT were 681×539 pixels in size as shown in Figure 3.

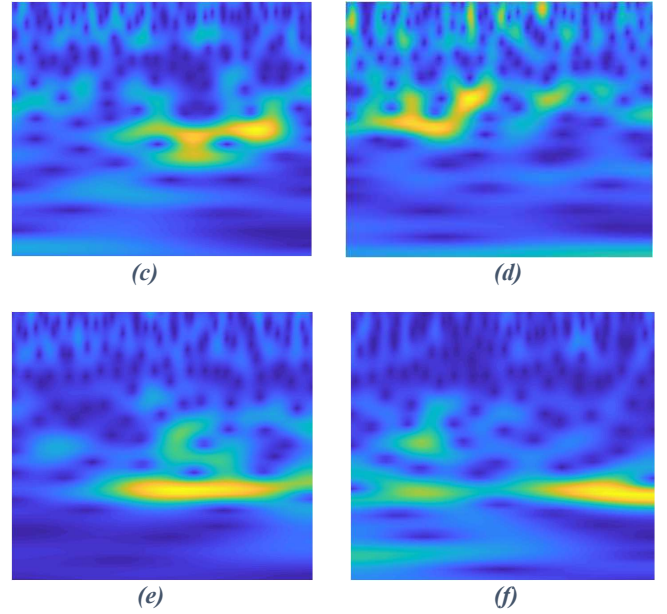
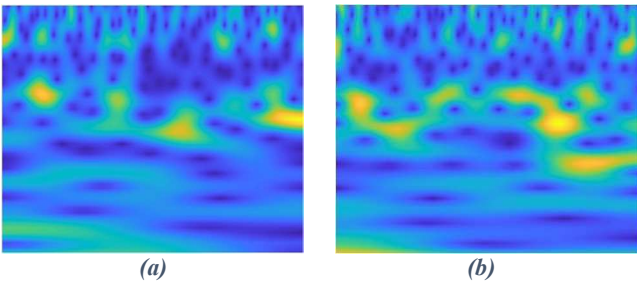


Figure 3. Scalograms obtained using CWT (a) (c) (e) shows C3,Cz,C4 electrode of Left MI respectively and (b)(d)(f) shows C3,Cz,C4 electrodes of Right MI respectively.

C. CLASSIFICATION

The retrieved features are given in the deep learning classification method as mentioned previously. CNNs are a type of neural network that has several layers having a fully connected layer in output and numerous convolution-pooling layer pairs. The typical CNN is designed to distinguish forms in images and is invariant in most cases based on location [7]. The input picture is composed of several 2D filters within the convolutional layer and then subsampled to a lower size inside the pooling layer. In order to reduce classification error, the back-propagation algorithm is used to learn network weights and filters in the convolution layer [8]. Our data, on the other hand, combines frequency, time, and electrode position. For classification performance, the vertical position of the activation on the input picture is important, although not its horizontal location. As a result, rather than the conventional 2D filtering, we provided filters with an analogous height because only the horizontal axis is used as input and 1D filtering [20].

D. Convolutional Neural Networks Architecture

We use a CNN version to enable MI task recognition. For 2D picture categorization, the traditional CNN has proved to be rather effective. Input, output, and numerous more hidden layers. A CNN is made up of convolutional-pooling layers in pairs and a fully connected layer. [13]. The conventional CNN takes the image's features from the convolutional layer's 2D kernel and subsamples it in the pooling layer to a lesser size. Within a fully linked layer, the reduced picture is then categorized. It should be noted that the MI EEG picture used as an input differs from the standard CNN input file format [14]. Our goal is to categorize hand MI activities using characteristics within the frequency represented on the vertical axis, in contrast to the time axis represented on the horizontal axis. To capture the characteristics of frequency and electrode position on the same time axis, we used a MI

classification technique based on CNN using 1D kernels rather than typical 2D kernels [15].

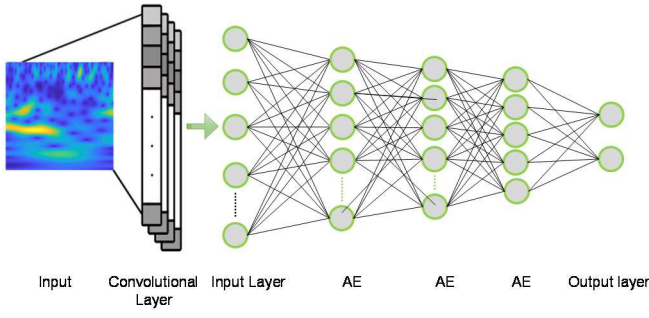


Figure 4 Typical Representation of CNN layers

The output of the $(N_t-2) \times 1$ convolutional layer is added to the input of the max-pooling layer where N_t is the filter kernel matrix of the $t \times t$ dimension. Max-pooling is performed with a 10-sampling point and no padding. As a result, the max-pooling layer subsamples the dimensions of the convolutional layer's output to three minus one dimension for 30 kernels. Finally, the completely linked layer is utilized for two kinds of MI jobs: left and right hand. The proposed CNN model is trained using labeled training data, and the classification error is calculated with the CNN output difference and target data difference. The back-propagation technique uses gradient descent to reduce mistakes and update the weights of the network. The performance of MI categorization in test data is computed using the network's learned weights [16].

E. Transfer Learning

An $M \times M \times D$ picture is fed into the CNN, with M denoting the image's width and height and D denoting the image's depth. The convolutional layers have k filters with sizes $N \times N \times Q$, which are smaller than the input image. The image's width and height can be adjusted. For the sake of simplicity, these values are maintained equally. For image identification, a well-known CNN network GoogLeNet, AlexNet, and ResNet50 is utilized [17]. The GoogLeNet architecture is made up of 170 layers of the CNN framework. AlexNet architecture is made up of 25 layers of the CNN framework. And ResNet50 is made up of 50 layers deep.

Each convolutional layer produces a collection of K_s , or $(M-N-1) \times (M-N-1)$ feature maps. Convoluting the input picture X with the weight filter W^k and adding the bias b_k as yielding one feature map h^k , given as.

$$h_{ij}^k = f((W^k * x) + b_k) \quad (2)$$

To decrease the dimension, a pooling layer is added to the output of the convolutional layer. In this layer, several pooling algorithms such as max-pooling and average-pooling can be employed. These techniques work on feature map sub-regions and return the maximum or average of those sub-regions [18]. Several fully connected layers might be added after convolutional-pooling layers. These layers are the same as in traditional feed-forward networks. Backpropagation algorithms, similar to MLP, may be used to train CNNs

IV. RESULTS

The tests were carried out in a MATLAB environment using an Intel 2.5 GHz Core i7 PC with 16 GB of RAM and 6 GB

of GPU in this study. The suggested networks were designed and tested using the MATLAB deep learning toolbox. In order to execute 1D convolution and max pooling in the CNN network, the toolbox was significantly modified.

It is long established that MI features are reflected in the mu-bands (8-13 Hz) and beta-bands(13-30hz) in EEG data [2]. A similar decrease of EEG signal mu and beta bands, called event-related desynchronization (ERD), is detected in the contralateral brain area when left and right motor motions are imagined. Furthermore, event-related synchronization is the phenomenon in which the EEG signal capacity is restored in these frequency bands following MI tasks.

ERD/ERS patterns are represented as a shift in power within the MI EEG signal in contrast to a reference period prior to the commencement of motor movement imagery. To begin, the dataset's average of each MI EEG channel is computed over all individuals and trials. The rate of change of power with respect to the reference signals is then used to calculate the ERD/ERS patterns.

In our proposed method the average power graphs for each trial and channel are acquired. The average power of the alpha and beta bands is obtained. This is because when a motor imagery job is pictured, the majority of the information resides in these bands. For the BCI task, we discovered substantially higher power reductions in the beta band (18–22 Hz).

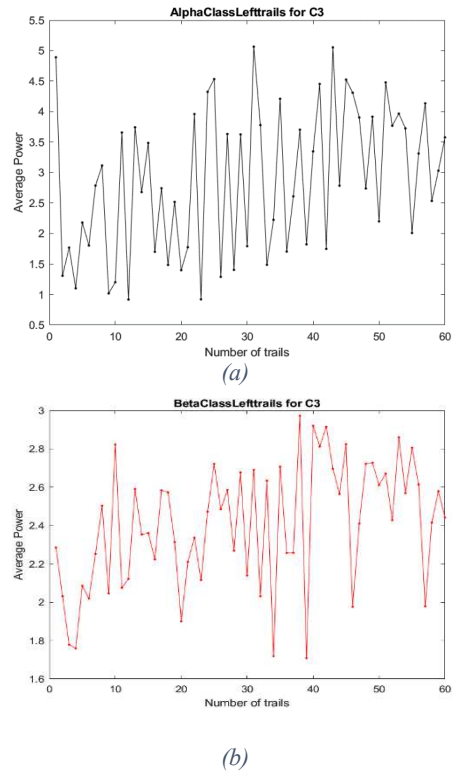


Figure 5 Power analysis (a) Alpha band left C3 (b) Beta band left C3

Table 2 Accuracy comparison among various different studies

	GoogLeNet	AlexNet	ResNet50
Accuracy	82.81%	90.11%	85.40%

We tested the suggested CNN's classification performance on 1800 pictures from both the Left and Right classes.

Because of its better time-frequency resolution. CWT improves the clarity of ERD patterns in the mu-band from an EEG input picture without sacrificing time or electrode-frequency information. The dataset was divided into training and test sets for each participant in order to assess its correctness. As a result, 85 percent of each subject's 120 trials were chosen at random as the training set and the rest as the test set. The suggested CNN was used to analyze the test sets. The GoogLeNet architecture is made up of 170 layers of the CNN framework. Two 224x224x3 input class folders were supplied to the input layer. A batch training approach was used to train CNN for 30 epochs with a Mini batch size of 128. The AlexNet architecture is made up of 25 layers of the CNN framework. Two 227x227x3 input class folders were supplied to the input layer. A batch training approach was used to train CNN for 30 epochs with a Mini batch size of 128. The ResNet50 architecture is made up of 50 layers of the CNN framework. Two 224x224x3 input class folders were supplied to the input layer. A batch training approach was used to train CNN for 30 epochs with a Mini batch size of 128.

The suggested approach with AlexNet achieves a training accuracy of 99 percent and a testing accuracy of 90.11 percent on average, with GoogLeNet suggested approach achieving a training accuracy of 89 percent and a testing accuracy of 82.81 percent, with ResNet50 suggested approach achieving a training accuracy 92.32 percent and 85.40 percent testing accuracy. This study showed that AlexNet performs well among other architectures with this dataset. For the categorization of MI tasks, the suggested technique provides an improvement.

V. CONCLUSION

In this research, we provide a replacement of continuous wavelet transform and convolutional neural network by employing transfer learning (GoogLeNet, AlexNet, ResNet50) based decoding technique, which is distinct from previously reported works, to recognize motor imagery tasks. The proposed technique consists of two stages: picture creation for motor imagery EEG data using the continuous wavelet transform and then the proposed one-dimensional convolutional neural network was used to classify motor imagery tasks. In addition to frequency and electrode placement, we confirm that EEG images contain event-related desynchronizing patterns in the resultant motor imagery. Following that, 3 convolutional neural networks in one dimension of 170,25,50 layers are constructed utilizing the motor imagery EEG picture as input to interpret the two separate motor imagery tasks. The network's goal is to

capture the one-dimensional dynamics of EEG picture input motor images.

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